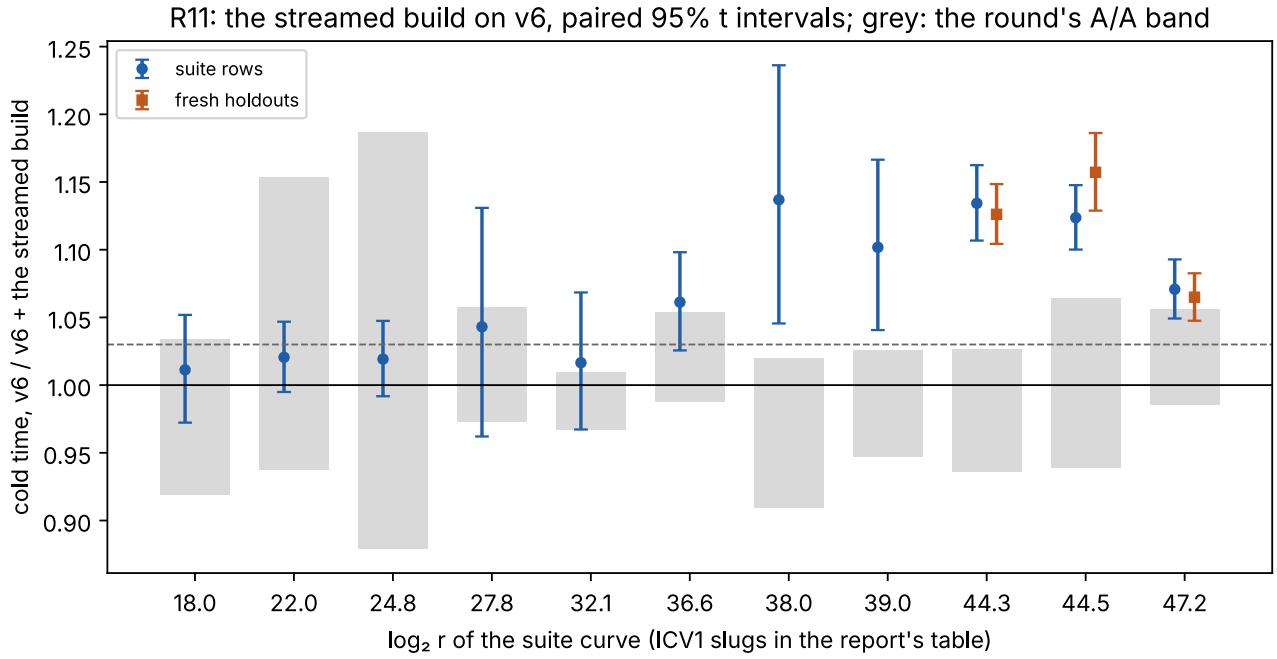
R11: the pair-table build by partition streams and per-run filters

Accepted, as baseline v7 (engineering). v6 (v4 plus R06′s scan key and R08′s folding subtraction) against v6 plus R11′s candidate (`candidate-v6.patch`: each pair's `(hash, word)` straight into one of 256 partitions on its run's stream, and a presence filter per run ORed at the end), both built with `rustc 1.97.0`, on an Emerald Rapids host of the reference class, 2026-10-16. Protocol: `research/ic_tool_program/rounds/R11-build-streams/PROTOCOL.md` (declaration, amendment 1); analysis: `analysis.json` (`icprog analyse r11`); run trees: `runs.tar.xz` (SHA-256 `5fe8adebbcc9e02f…`).

- **Tests:** 207 of 207 `koblitz_fast` and `koblitz_index_calculus` tests on the candidate's tree, among them the word-for-word table test at one, two, three and five threads.
- **Pin:** the candidate equals v0′s outputs on all 90 suite rows.
- **Timed:** 946 isolated processes (A/A 224, suite 401, holdouts 321), 6 contended and re-run, none failed; the driver task was relaunched once on the same container and resumed from disk.
- **Decision:** at each target size the cold interval lies above 1.03 on the suite rows and on the fresh holdouts; no size regresses beyond its A/A band. Predicted 1.08–1.12, 1.08–1.12 and 1.04–1.07.
- **The build phase** (medians over every repetition, base → candidate): 187.1 → 124.5 ms, 190.4 → 126.3 ms and 331.1 → 219.0 ms, 1.50–1.51× faster. Collection is unchanged (0.98–1.01).



research/ic_tool_program/rounds/R11-build-streams/analysis.json; 946 isolated processes, 6 contended (excluded); dashed: the 1.03 acceptance line

Figure 1: Cold time, v6 over the candidate (above 1, the candidate is faster), paired 95% t intervals; grey: the round's own A/A band; dashed: 1.03. Values from `analysis.json`.

Suite rows (five rounds)

curve	$\log_2$ r	rows	cold [95%]	collection	S cold, v6 → v7	A/A band
`icv1-f2m19-tm797-9c54981b`	18.002	2	1.011 [0.972, 1.052]	0.795	124.09 → 122.43	0.919–1.034
`icv1-f2m23-tm5197-1f85e9e1`	22.001	2	1.021 [0.995, 1.047]	1.005	32.33 → 31.42	0.938–1.154
`icv1-f2m45-tm6236725-40939294`	24.803	2	1.019 [0.992, 1.047]	0.982	19.77 → 19.38	0.879–1.187
`icv1-f2m37-tm534059-32aad96b`	27.781	2	1.043 [0.962, 1.131]	1.015	8.91 → 8.44	0.973–1.057
`icv1-f2m43-tm998717-e2e742b0`	32.113	2	1.017 [0.967, 1.068]	0.999	4.57 → 4.5	0.968–1.01
`icv1-f2m47-t22705043-f4e44623`	36.636	2	1.061 [1.026, 1.098]	1.003	2.53 → 2.4	0.988–1.054
`icv1-f2m57-tm747311035-c1f545af`	38.002	2	1.137 [1.046, 1.236]	1.049	2.91 → 2.62	0.909–1.02
`icv1-f2m41-tm2308219-7f48b14a`	39.0	2	1.102 [1.041, 1.166]	0.981	3.38 → 3.1	0.947–1.026
`icv1-f2m53-tm56619371-dac20a85`	44.259	8	1.134 [1.107, 1.162]	1.014	3.87 → 3.48	0.936–1.026
`icv1-f2m59-tm943548413-98844ecc`	44.517	8	1.124 [1.1, 1.148]	0.982	3.53 → 3.11	0.939–1.064
`icv1-f2m61-t158598901-ab42b6c5`	47.211	8	1.071 [1.049, 1.093]	1.011	4.82 → 4.51	0.986–1.056

Fresh holdouts (seeds 234–237, T159–T166; ten rounds at 2^44.3)

curve	$\log_2$ r	pairs	cold [95%]	S cold, v6 → v7
`icv1-f2m53-tm56619371-dac20a85`	44.259	80	1.126 [1.104, 1.148]	3.83 → 3.38

`icv1-f2m59-tm943548413-98844ecc`	44.517	40	1.157 [1.129, 1.186]	3.72 ⟶ 3.15
`icv1-f2m61-t158598901-ab42b6c5`	47.211	40	1.065 [1.048, 1.083]	4.57 ⟶ 4.26

Scope

One host and one core, the reference class. The table is word for word the same, by construction and by test, so the counts and the ratio to the floor are unchanged. Chained with R06 and R08, cold time is 1.80–1.86× faster than v4 at the three target sizes (a product of paired ratios, not a measurement). The distance to matched rho is the rule at v6′s until a rule runs at v7.